11. The average percentage of failure in a certain examination is 40. The probability that out of a group of 6 candidates atleast 4 passed in the examination are

(a) 0.5443 (b) 0.4543

(c) 0.5543 (d) 0.4573

12. Forty percent of the passengers who fly on a certain route do not check in any luggage. The planes on this route seat 15 passengers. For a full flight, what is the mean of the number of passengers who do not check in any luggage?

(a) 6.00 (b) 6.45

(c) 7.20 (d) 7.50

13. Which of the following statements is/are true regarding the normal distribution curve?

(a) it is symmetrical and bell shaped curve

(b) it is asymptotic in that each end approaches the horizontal axis but never reaches it

(c) its mean, median and mode are located at the same point

(d) all of the above statements are true.

14. Which of the following cannot generate a Poisson distribution?

(a) The number of telephone calls received in a ten-minute interval

(b) The number of customers arriving at a petrol station

(c) The number of bacteria found in a cubic feet of soil

(d) The number of misprints per page

15. The random variable X is normally distributed with a mean of 70 and a standard deviation of 10. What is the probability that X is between 72 and 84?

(a) 0.683 (b) 0.954

(c) 0.271 (d) 0.340

16. The starting annual salaries of newly qualified chartered accountants (CA's) in South Africa follow a normal distribution with a mean of ₹1,80,000 and a standard deviation of ₹10,000. What is the probability that a randomly selected newly qualified CA will earn between ₹1,65,000 and ₹1,75,000 per annum?

(a) 0.819 (b) 0.242

(c) 0.286 (d) 0.533

17. In a large statistics class the heights of the students are normally distributed with a mean of 172cm and a variance of 25cm. What proportion of students are between 165cm and 181cm in height?

(a) 0.954 (b) 0.601

(c) 0.718 (d) 0.883

18. A statistical analysis of long-distance telephone calls indicates that the length of these calls is normally distributed with a mean of 240 seconds and a standard deviation of 40 seconds. What proportion of calls lasts less than 180 seconds?

(a) 0.214 (b) 0.094

(c) 0933 (d) 0.067

19. Cape town is estimated to have 21% of homes whose owners subscribe to the satelite service, DSTV. If a random sample of your home in taken, what is the probability that all four home subscribe to DSTV?

(a) 0.2100 (b) 0.5000

(c) 0.8791 (d) 0.0019

20. Using the standard normal table, the sum of the probabilities to the right of z = 2.18 and to the left of z = –1.75 is

(a) 0.4854 (b) 0.4599

(c) 0.0146 (d) 0.0547

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